DA5000W Linear algebra and optimization

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Tutorial 3 Linear algebra

1 Theoretical Recap: Eigenvalues, Eigenvectors, and Diagonalization

1.1 Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix over a field $\mathbb{F}$. A nonzero vector $v \in \mathbb{F}^n$ is called an *eigenvector* of A corresponding to the scalar $\lambda \in \mathbb{F}$ if

$$Av = \lambda v.$$

The scalar λ is called an *eigenvalue* of A.

1.2 Characteristic Polynomial

For an $n \times n$ matrix A, the *characteristic polynomial* of A is defined by

$$p_A(\lambda) = \det(A - \lambda I).$$

The roots of $p_A(\lambda) = 0$ are the eigenvalues of A.

A has atmost n distinct eigenvalues.

1.3 Eigenspace

For each eigenvalue λ of $A \in M_{n\times n}(F)$, the set of all eigenvectors corresponding to λ, together with the zero vector, forms a subspace of $\mathbb{F}^n$, called the *eigenspace* of A corresponding to λ:

$$E_\lambda = \{v \in \mathbb{F}^n \mid (A - \lambda I)v = 0\}.$$

1.4 Algebraic and Geometric Multiplicities

Let λ be an eigenvalue of $A \in M_{n\times n}(F)$.

- The *algebraic multiplicity* of λ is the largest positive integer k for which $(t - \lambda)^k$ is a factor of $f(t)$.
- The *Geometric Multiplicity* of λ, denoted by $m_g(\lambda)$, is the dimension of the eigenspace corresponding to λ:

  $$m_g(\lambda) = \dim\big(\mathrm{Null}(A - \lambda I)\big).$$

  Equivalently, it is the number of linearly independent eigenvectors associated with λ.

For every eigenvalue λ,

$$1 \leq \text{geometric multiplicity of } \lambda \leq \text{algebraic multiplicity of } \lambda.$$

1.5 Diagonalization Criterion

A matrix $A \in \mathbb{F}^{n\times n}$ is *diagonalizable* if and only if there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

Equivalently, A is diagonalizable if and only if the sum of the dimensions of all its eigenspaces equals n, that is, if A has n linearly independent eigenvectors.

To diagonalize a matrix or a linear operator is to find a basis of eigenvectors and the corresponding eigenvalues.

1.6 Procedure to Diagonalize a Matrix

To diagonalize a square matrix A:

1. Compute the characteristic polynomial $\det(A - \lambda I) = 0$.
2. Find all eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$.
3. For each λ_i, find the eigenspace $E_{\lambda_i} = \ker(A - \lambda_i I)$.
4. Collect a basis of eigenvectors from all eigenspaces.
5. If the total number of independent eigenvectors is n, form the matrix $P = [v_1\ v_2\ \ldots\ v_n]$ and the diagonal matrix $D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$, satisfying $A = PDP^{-1}$.

1.7 Spectral Theorem (for Real Symmetric Matrices)

If A is a real symmetric matrix, then:

1. All eigenvalues of A are real.
2. Eigenvectors corresponding to distinct eigenvalues are orthogonal.
3. There exists an orthogonal matrix Q such that

$$Q^T AQ = D,$$

where D is diagonal.

Hence, every real symmetric matrix is orthogonally diagonalizable.

1.8 Theorem

Let T be a linear operator on a vector space V, and let $\lambda_1, \lambda_2, \cdots, \lambda_n$ be distinct eigenvalues of T.

If $v_1, v_2, \cdots, v_k$ are eigenvectors of T such that λ_i corresponds to $v_i (1 \le i \le k)$, then $v_1, v_2, ..., v_k$ is linearly independent.

Corollary Let T be a linear operator on an n-dimensional vector space V. If T has n distinct eigenvalues, then T is diagonalizable.

2 Example: Finding Eigenvalues and Eigenvectors of a 3×3 Matrix

Let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}.$$

We will find the eigenvalues and eigenvectors of A.

2.1 Step 1: Characteristic Equation

Eigenvalues λ satisfy

$$\det(A - \lambda I) = 0.$$

Hence,

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & 4 \\ 0 & 4 & 9-\lambda \end{bmatrix}.$$

The determinant is

$$\det(A - \lambda I) = (2-\lambda) \begin{vmatrix} 3-\lambda & 4 \\ 4 & 9-\lambda \end{vmatrix} = (2-\lambda)[(3-\lambda)(9-\lambda) - 16].$$

Expanding,

$$(2-\lambda)(\lambda^2 - 12\lambda + 11) = 0.$$

2.2 Step 2: Finding Eigenvalues

Solve $\lambda^2 - 12\lambda + 11 = 0$:

$$\lambda = \frac{12 \pm \sqrt{12^2 - 4(1)(11)}}{2} = \frac{12 \pm \sqrt{100}}{2} = \frac{12 \pm 10}{2}.$$

Thus,

$$\lambda_1 = 11, \quad \lambda_2 = 1.$$

Including the first factor $(\lambda - 2) = 0$, we have

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_3 = 11.}$$

2.3 Step 3: Finding Eigenvectors

We solve $(A - \lambda I)x = 0$ for each λ.

(a) For $\lambda = 2$:

$$A - 2I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 4 & 7 \end{bmatrix}.$$

The equations are

$$y + 4z = 0, \quad 4y + 7z = 0.$$

From the first, $y = -4z$. Substituting gives $-9z = 0 \Rightarrow z = 0$, hence $y = 0$ and x is free.

$$\Rightarrow v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

(b) For $\lambda = 1$:

$$A - I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 4 & 8 \end{bmatrix}.$$

Equations:

$$x = 0, \quad 2y + 4z = 0 \Rightarrow y = -2z.$$

Let $z = 1$:

$$\Rightarrow v_2 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}.$$

(c) For $\lambda = 11$:

$$A - 11I = \begin{bmatrix} -9 & 0 & 0 \\ 0 & -8 & 4 \\ 0 & 4 & -2 \end{bmatrix}.$$

Equations:

$$x = 0, \quad -8y + 4z = 0 \Rightarrow y = \frac{1}{2}z.$$

Let $z = 2$:

$$\Rightarrow v_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

2.4 Diagonalization of A

Let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}, \qquad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

The diagonal matrix of eigenvalues (in the same column order) is

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 11 \end{bmatrix}.$$

One computes (or verifies) the inverse

$$P^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{2}{5} \end{bmatrix},$$

and checks

$$P^{-1}AP = D.$$

Equivalently, A is diagonalizable and can be written as

$$\boxed{A = PDP^{-1}}.$$

3 Singular Value Decomposition (SVD)

3.1 Singular Value Decomposition

Let A be any real $m \times n$ matrix. Then there exist orthogonal matrices $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$, and a diagonal matrix $\Sigma \in \mathbb{R}^{m\times n}$ such that

$$A = U\Sigma V^T.$$

Here,

- The diagonal entries $\sigma_1, \sigma_2, \ldots, \sigma_r$ of Σ are called the *singular values* of A, where $r = \text{rank}(A)$.
- The columns of $U \in \mathbb{R}^{m\times r}$ are called the *left singular vectors* of $A \in \mathbb{R}^{m\times n}$.
- The columns of $V \in \mathbb{R}^{n\times r}$ are called the *right singular vectors* of $A \in \mathbb{R}^{m\times n}$.

3.2 Computation of the SVD

The SVD of A is obtained as follows:

1. Compute A^TA and find its eigenvalues λ_i and eigenvectors v_i.
2. The singular values are $\sigma_i = \sqrt{\lambda_i}$.
3. The right singular vectors form $V = [v_1, v_2, \ldots, v_n]$.

4. The left singular vectors are computed as

$$u_i = \frac{1}{\sigma_i} A v_i, \quad \text{for each nonzero } \sigma_i.$$

Finally,

$$A = U \Sigma V^T,$$

where Σ contains the singular values along the diagonal.

3.3 Example: Computing the SVD of a 2×2 Matrix

Let

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}.$$

Step 1: Compute $A^T A$.

$$A^T A = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}.$$

Step 2: Find right singular vectors. For $\lambda_1 = 7 + \sqrt{13}$,

$$(A^T A - \lambda_1 I) = \begin{bmatrix} 2 - \sqrt{13} & 3 \\ 3 & -2 - \sqrt{13} \end{bmatrix}.$$

This gives the eigenvector (up to scaling)

$$v_1 = \begin{bmatrix} 3 \\ \sqrt{13} - 2 \end{bmatrix}.$$

Normalize v_1:

$$v_1 = \frac{1}{\sqrt{9 + (\sqrt{13} - 2)^2}} \begin{bmatrix} 3 \\ \sqrt{13} - 2 \end{bmatrix} = \frac{1}{\sqrt{18 - 4\sqrt{13}}} \begin{bmatrix} 3 \\ \sqrt{13} - 2 \end{bmatrix}.$$

Now choose v_2 orthogonal to v_1:

$$v_2 = \frac{1}{\sqrt{18 + 4\sqrt{13}}} \begin{bmatrix} 2 + \sqrt{13} \\ -3 \end{bmatrix}.$$

Hence,

$$V = [v_1 \; v_2].$$

Step 3: Find singular values.

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{7 + \sqrt{13}}, \qquad \sigma_2 = \sqrt{\lambda_2} = \sqrt{7 - \sqrt{13}}.$$

Step 4: Find left singular vectors.

$$u_i = \frac{1}{\sigma_i} A v_i.$$

Hence,

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{\sqrt{7 + \sqrt{13}}} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} v_1, \quad u_2 = \frac{1}{\sigma_2} A v_2 = \frac{1}{\sqrt{7 - \sqrt{13}}} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} v_2.$$

After normalization,

$$U = [u_1 \; u_2].$$

Therefore, the singular value decomposition is

$$A = U \Sigma V^T, \quad \Sigma = \begin{bmatrix} \sqrt{7 + \sqrt{13}} & 0 \\ 0 & \sqrt{7 - \sqrt{13}} \end{bmatrix}.$$

Step 6: Form the decomposition.

$$A = U\Sigma V^T, \quad \Sigma = \begin{bmatrix} 3.61 & 0 \\ 0 & 1.66 \end{bmatrix}.$$

Interpretation

The SVD expresses A as a product of:

$$A = U\Sigma V^T,$$

where:

- V rotates the input space (right singular vectors),
- Σ scales along principal axes (singular values),
- U rotates to the output space (left singular vectors).

This decomposition is widely used in data compression, PCA, and noise reduction.